Comparative Study of AI-Powered Interview and Evaluation Systems: Speech-to-text Recognition

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Abstract : **AI-powered interviewing and speech recognition technologies have become more popular in recent years, which are used in a variety of sectors, including accessibility, education and hiring processes. The goal of text-to-speech (TTS) recognition is to translate spoken language into text using techniques like Artificial Neural Networks (ANN), Hidden Markov Models (HMM) and deep learning-based strategies like transformer topologies and Long Short-Term Memory (LSTM) networks. On the other hand, AI-driven online interview platforms use speech processing, facial expression identification, sentiment analysis as well as Natural Language Processing (NLP) to evaluate candidates' expertise, confidence level and emotional approach during virtual interviews. This study highlights the technological underpinnings, constraints and benefits of the approaches utilised in AI-driven interview systems and TTS recognition. This work suggests a speech-to-text system driven by Artificial Intelligence(AI) that will simplify and standardize online interview assessments to enhance the chances of selection. Utilising developments in natural language processing (NLP), machine learning and speech recognition. The system automatically transcribes applicant responses and assesses them based on predetermined standards, such as communication skills, subject knowledge, problem-solving ability etc. Through the reduction of human subjectivity, the technology guarantees efficiency, fairness and scalability, allowing organizations to manage a large number of interviews while offering data-driven insights for well-informed decision making for the recruitment processes.**

Keywords : Speech-to-text, Computer vision, NLP, Data driven insights, Machine learning, Artificial intelligence, Evaluation, Interview, Sentiment analysis, Recruitment, Accuracy, Hiring, Evaluation .

I. Introduction

As the world progresses forward, recruitment becomes one of the most important processes. Not only this, but it is also one of the most difficult as efficiency, accuracy, and fairness all play their individual roles in the process of recruitment. An organization or a candidate each have their own needs which make the perfect employment strategy unique. A perfect solution is not easy to find due to the interviewing systems' incoherent nature. Due to traditional interviewing techniques, subjectivity, inconsistency, and general inefficiency make it hard to pinpoint a candidate's actual potential. Thankfully, the integration of AI recruitment offers a solution to these challenges, enabling accurate, unbiased, and efficient evaluation of candidates.

An AI-based system is suggested in this study to resolve the human subjectivity gap between decision-making and data analysis. The current approach integrates computer vision and machine learning technology, which facilitates the extraction of relevant information from the videos and audios of the candidates. The assessment system calculates the emotional reaction, level of understanding, and the confidence of respondents during the interviewing process. These features, or attributes, are critical in forecasting the organizational value that the candidates will greatly contribute.

This research consists of critical review and evaluation of system concepts, which will enhance the recruitment evaluation processes through automation of manual tasks and elimination of human contact and biases. The system is designed to support self-service, operational and cost effectiveness, and compliance to legal and ethical standards by providing solutions to evaluation biases and systematic distortions due to unfounded negative ratings and unqualified review malpractice. Highly sophisticated AI systems perform human reasoning skills to solve the issue posed to this study. Deep learning models for recognition of emotions that are manifested through facial expressions make it possible to capture very delicate pieces of information. At the same time, Natural Language Processing tools identify the intelligibility and logic completeness of the speech given by the person under assessment. Enable companies to use knowledge-based assessment technology.

II. LITERATURE SURVEY

The field of Artificial Intelligence is evolving at an unprecedented pace and has had measurable impacts in the recruitment space, just like every other industry. The development of AI-powered interview systems has been made possible by the tremendous growth in automated speech recognition technologies. Research study in this domain of vast AI-field reveals the use of advanced architecture of Artificial Intelligence such as Recurrent Neural Networks (RNNs), Convolutional Neural Networks (CNN), and Transformers neural networks which are capable of accurately transcribing candidates' utterances in real time irrespective of background noise or speaker accent[1][6]. With these capabilities, recruiting companies can track and evaluate candidate responses without using resources, time, and manual methods, which in most cases lead to malpractice and errors.

Besides speech recognition, Natural Language Processing (NLP) has significantly automated the processing and evaluation of the written text. NLP-driven systems automatically capture the fundamental meaning of a user's input in order to judge his or her domain knowledge, problem solving ability, and communication skills as well as self-esteem indicators [2]. These systems employ sentiment analysis, syntactic parsing and semantic role labeling to make the assessment of the given answer and its intent verifiable and of high quality, which makes sure that evaluations remain consistent across candidates [2][7]. NLP systems determine the domain knowledge, problem-solving skills, and communication proficiency of a candidate using advanced linguistic and semantic feature extraction [2]. To maintain objectivity in evaluating quality and intent of responses across all candidates, these systems employ sentiment analysis, syntactic parsing, and semantic role labeling when responding [2][7].

AI-driven recruitment platforms enhance their overall value by addressing challenges associated with remote hiring that have recently emerged. Pre-screening interviews and technical assessments are conducted automatically, resulting in a shorter time the overall recruiting process takes and increased scalability for businesses with large numbers of applicants. AI systems for virtual interviewing have been shown to facilitate recruitment as presented in other studies, particularly in contexts where decisions require a high degree of precision and impartiality [3][4]. Apart from easing participation from different locations, these systems also provide comprehensive data-driven reports to aid recruiters, thus increasing the efficacy of attracting skilled professionals [3].

One more dimension in which AI works toward eliminating inefficiencies is human bias, which has been a challenge in most hiring processes. These systems make it possible to evaluate candidates against objective parameters, automating the process and guaranteeing all candidates receive equal opportunities. This corresponds very well to technical positions where the unbiased assessment of technical and problem-solving abilities is the most important [4].

Together with correction of inefficiencies, AI in recruiting undertakes the human bias factor, which has been a problem in systematic hiring processes. These systems, through automation of candidate evaluation using set criteria, enable equity and diversity within the recruitment process. AI systems have proven effective in some technical positions where there is a need for objective assessment of the technical and problem solving skills of the applicants[4]. This guarantees that the candidates will be evaluated only based on their performance, which is free from subjectivity or any self-implicit bias of human judgment. AI-based mock interview systems also proved useful in preparing candidates for actual interviews. These systems run mock interviews based on which they provide feedback that cover self-esteem, emotional condition, and content performance. Such systems not only assist candidates in improving their readiness but also enhance their chances of success in actual interviews [5][7]. The paper focuses on leveraging machine learning, computer vision, and natural language processing for objective candidate assessment. Studies have demonstrated the effectiveness of emotion recognition through facial expression analysis and confidence detection using speech patterns. Various approaches combine neural networks and AI-driven chatbots to enhance the interactive interview process and provide unbiased, data-driven insights [6].Emotional intelligence and confidence detection systems, in particular, utilize computer vision and NLP models to gauge candidates' emotional states during interactions, providing a holistic evaluation of their suitability for the role [7].

Such systems aid candidates in addressing their preparations and increase the likelihood of succeeding in real interviews. The paper aims at machine learning driven computer vision and natural language processing applications for objective evaluation of candidates. Research has validated emotion recognition from facial analysis and confidence analysis from voice signals. Different methods combine neural networks with AI chatbots for automated active interviewing and impartial, evidence-based assessments. Systems for emotional intelligence and confidence estimation, in particular, apply computer vision and natural language processing models to evaluate the emotions of respondents during the interview and provide overall measure of the candidate's fitness for the job.

Comparative studies of speech-to-text conversion methodologies have also been crucial in identifying the most efficient techniques for various applications. These studies evaluate models like Hidden Markov Models (HMMs), Long Short-Term Memory (LSTM) networks, and hybrid systems to determine their efficacy in transcribing diverse linguistic datasets. Results show that AI techniques tailored to specific dialects and languages perform significantly better, ensuring inclusivity in recruitment practices [8].

Lastly, the combination of speech recognition and NLP (Natural Language Processing) with AI-driven evaluation frameworks provides recruiters with detailed, data-driven analyses of candidate’s performance. These systems not only streamline the hiring process but also ensure the confidentiality, privacy and security of candidate’s data, considering key concerns and issues in modern hiring practices. By offering scalable, fair and unbiased solutions for recruitment processes, AI-powered systems bridge the gap between recruiters and candidates, ensuring efficiency as well as transparency in talent acquisition.

III. METHODOLOGY

The methodology for developing an AI-powered system for efficient and unbiased online interview evaluation, which integrates advanced technologies. Such as speech recognition, Natural Language Processing (NLP), Deep Learning and Machine Learning. These systematic approaches ensure that the recruitment process is not only efficient but also scalable and transparent. The methodology focuses on real-time audio and video analysis. Also the system helps to assess the emotion and confidence of a candidate. The data-driven evaluation of the interview to generate detailed feedback with various evaluation factors as well as evaluation for both recruiters and candidates. The system design prioritizes user experience, user interface, scalability and ethical considerations, ensuring a seamless and fair recruitment process.

Approach:

This method is flexible and precise, making it modular and focused on the details for any recruitment effort. Initially, users are recorded on audio and video, which is followed by filtering and normalization preprocesses. Speech to text processes like RNNs and transformers are then used to transform speech to text, followed by natural language processing that performs sentiment, domain knowledge, and communication effectiveness analysis. At the same time, the facial video data is analyzed for engagement and confidence through facial emotion recognition.While the insights gathered by the system over this procedure are scored based on unemotional and logical criteria, their accuracy being cross-verified at every stage, The reports, the recruiters get, are helpful and The system continuously improves itself by using the user's information and progress in machine learning.

This system suggests that the heuristic processes of candidate evaluation need to be automated, as they may impede the recruiters' ability to obtain accurate information. The methodology is outlined in the following sections:

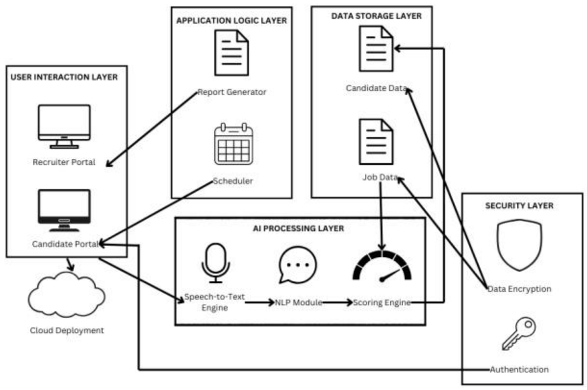

Fig1: System Architecture Diagram

1. **Candidate Interaction and Data Collection**

The system begins when candidates connect to the platform via a user-friendly well designed web interface. Audio and video inputs are recorded in real time during the interview process. The microphone records language, and the webcam records facial expressions and body language, ensuring comprehensive data collection for further evaluation.

2. **Data Preprocessing**

The collected data undergoes preprocessing to prepare it for further analysis and evaluation:

- **Audio Preprocessing:** Background noise is filtered, and audio levels are normalized. The processed audio is then converted to text using advanced speech-to-text algorithms, such as transformer models or Recurrent Neural Networks (RNNs).
- **Video Preprocessing:** Facial features are analyzed using models like Convolutional Neural Networks (CNNs) or Haar Cascades to detect engagement, emotional states, and confidence levels.

3. **Speech-to-Text Conversion**

The speech input is transcribed into text using state-of-the-art speech recognition technologies. These technologies include Hidden Markov Models (HMMs), RNNs, or transformer-based models, ensuring accurate transcription even in diverse linguistic contexts.

4. **Natural Language Processing (NLP) Analysis**

The transcribed text is further analyzed using NLP techniques to assess the following:

- **Sentiment and Tone Analysis:** Determines the candidate's emotional state during responses.
- **Keyword Extraction:** Identifies domain-specific terms to evaluate subject matter knowledge.
- **Response Quality:** Analyzes the logical flow, coherence, and clarity of answers.

5. **Feature Extraction**

Key features are extracted from the audio, video, and text inputs to form the foundation for evaluation:

- **Emotion and Confidence Metrics:** Derived from voice modulation, pauses, and facial expressions.
- **Technical Knowledge:** Measured by the presence and relevance of domain-specific terms.
- **Analytical Thinking:** Evaluated by the structure and logic in problem-solving explanations.

6. **Scoring and Evaluation**

The system evaluates candidates across predefined dimensions, assigning weighted scores to each category:

- **Problem-Solving Skills (40%)**
- **Communication Skills (30%)**
- **Technical Expertise (30%)**

The weighted scores are combined to produce a comprehensive assessment, ensuring a balanced evaluation.

7. **Report Generation**

The systems can also generate a detailed, data-driven report for each candidate and they can download it. Reports include:

7.1 Quantitative Scores: The evaluation metrics are calculated based on weighted scores for predefined dimensions. The final score is computed as:

$$S_{final} = \text{w1. } S_{problem-solving} + \text{w2. } S_{communication} + \text{w3. } S_{technical}$$

Where:

w1, w2, w3 are the weights assigned to each metric

$S_{problem-solving}$, $S_{communication}$, $S_{technical}$ are the individual scores for each dimension.

The scores are derived as follows:

- **Problem-Solving Skills:** Assessed using structured logic and analytical flow in responses.
- **Communication Skills:** Evaluated based on clarity, coherence, and engagement during responses.
- **Technical Expertise:** Measured by the use of domain-specific terms and the relevance of answers.

BLEU Score (Bilingual Evaluation Understudy)

Assesses the quality of generated text against a reference.

$$BLEU = BP.exp(\sum_{n=1}^{N} w_n.log\ p_n)$$

BP: Brevity penalty for overly short translations.

p_n : Precision for n-grams (e.g., uni-grams, bi-grams).

w_n : Weight for each n-gram precision

Qualitative Insights: Summaries of strengths, weaknesses, and performance areas.

a) Emotion Recognition Precision and Recall

Assesses how well the system identifies candidate emotions.

$$Precision = \frac{True\ Positive}{(True\ Positive\ + False\ Positives)}$$

$$Recall = \frac{True\ Positive}{(True\ Positive\ + False\ Negatives)}$$

- **Recommendations:** Suggestions for improvement tailored to the candidate.

8. Recruiter Dashboard and Decision-Making

Recruiters access a centralized dashboard that provides a comparison of candidate reports, enabling informed decision-making. The visual interface allows for filtering and sorting of candidates based on scores and evaluation criteria.

9. Feedback Loop for Continuous Improvement

The system employs a feedback loop, where historical data from completed interviews is used to train machine learning algorithms. This iterative process refines the system's performance, ensuring increased accuracy and adaptability to diverse candidate profiles.

Paper	Tech Stack	Algorithm Used	Observation
Automated Video Interview Interface	Python, Flask, OpenCV, Google Speech API, TensorFlow, MongoDB, Librosa	HaarCascade for face detection, Random Forest for confidence prediction, DeepFace	Comprehensive multimodal approach with emotion recognition, speech-to-text, and confidence evaluation. Suitable for scalable and real-time interview systems.
NLP-Based Interview Assessment System	Python, NLTK, spaCy, Flask	TF-IDF for feature extraction, LDA for topic modeling, Random Forest for analysis	Focused primarily on text-based analysis with keyword extraction and grammar checks. Lacks video and audio processing, making it limited for multimodal assessments.
Predicting Personality Using Open-Ended Questions	Python, TF-IDF, LDA, Word2Vec, Doc2Vec	TF-IDF and LDA for personality prediction, Random Forest for regression modeling	Excels in personality prediction using open-ended questions. Strong NLP component for extracting key features, but lacks video/audio integration. Ideal for enhancing the NLP aspect of the system.
Speech Recognition Using Neural Networks	Python, TensorFlow, OpenCV, Recorder.js	Artificial Neural Networks (ANN), Backpropagation, Genetic Algorithms	Strong focus on speech recognition using neural networks. Offers robustness in phoneme detection but lacks emotional and NLP analysis components.

AI-Based Interview Evaluator	Python, Flask, OpenCV, Google Speech API, TensorFlow, Librosa, Keras, MongoDB	DeepFace for emotion detection, HaarCascade for facial recognition, Random Forest	Combines video, audio, and text processing for a holistic candidate evaluation. Offers chatbot integration for interactive experiences. Suitable for real-world deployment with minor NLP enhancements.

Table 1: Comparative Study Table

Analysis :

The study demonstrates a wide variety of approaches and tools used in the creation of AI-powered interviewing systems. Methods that use multimodal analysis which includes analysing texts, audios and videos. This shows more promise for thorough candidate assessment, taking into account important factors such as knowledge, candidates's confidence and emotions. Effective methods for building reliable systems include Random Forest Classifiers for assessing confidence, DeepFace for recognising emotions, and HaarCascade for facial detection.Similar to this, models that use neural networks for voice recognition show remarkable power in processing audio, but for comprehensive evaluations, NLP and emotion detection frameworks must be included.

The combination of BERT for textual analysis, DeepFace for emotion recognition, and Random Forest for confidence evaluation forms the core of a powerful and scalable interview evaluation system. Integrating these with a multimodal fusion model can ensure comprehensive assessments across multiple dimensions, making the system insightful, unbiased, and adaptable to diverse candidate pools.

IV. CONCLUSION

This research put forward a novel framework for the assessment of recruitment reviews through artificial intelligence. It leverages advanced speech recognition, NLP, machine learning, and other remote configuration integration technologies to automate work and make it more efficient, scalable, and equitable. The system performs several tasks simultaneously – it automates language to text transcription, natural language processing, and evaluation of a candidate – eliminating human bias and providing comprehensive data control reports and standardized reviews. This guarantees the most unbiased assessment, improves evaluation accuracy and speed of candidate selection, and ensures completeness of analysis. In the future, emotion detection and evaluation metric improvement, as well as adaptive system feedback loop integration will be the primary focus. With the implementation of such artificial intelligent control tools, modern recruitment processes can be transformed completely, obtaining operational and strategic advantages that are crucial for organizations.

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